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FEDERAL SPECIFICATION

ADHESIVE, RUBBER BASE, (IN PRESSURIZED DISPENSERS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 This specification covers two types of rubber base adhesive which is relatively colorless, fast drying, and packaged in a pressurized container commonly known as an aerosol dispenser

1.2 Classification The rubber base adhesive shall be of the following types, as specified:

Type I - Regular (high solids).
Type II - Non-flammable (low solids).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein

Federal Specifications:

QQ-S-775 - Steel Sheets, Carbon, Zinc-Coated.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Shipping, Fiberboard, Wood-Cleated
PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-wall
PPP-F-320 - Fiberboard Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

Federal Standards

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies)
Fed Test Method Std No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8040

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2.2 Other publications The following documents form a part of this specification to the extent specified herein Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply

American Society for Testing and Materials (ASTM) Standards:

D 230-44 - Numbered Cotton Duck and Army Duck.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

Chemical Specialities Manufacturers Association (CSMA) Publications.

CSMA - Aerosol Guide - Vacuum Distillation:

Method for Determination of Volatile - Nonvolatile Ratios of Aerosol Formulations, Vacuum Distillation Method (Adopted December 4, 1956) (Fifth Edition, March 1966).

CSMA - Aerosol Guide - Flammability:

Revised Flammability Test Methods for Aerosol Products (adopted September 12, 1963). (Fifth Edition, March 1966).

(Application for copies should be addressed to Chemical Specialities Manufacturers Association, Inc. Suite 1120. 1001 Connecticut Avenue, N. W , Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606)

National Motor Freight Traffic Association Inc., Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3 1 Materials.

3.1.1 Type I adhesive shall be an elastomeric material with a hydrocarbon propellant and suitable solvent as required.

3.1.2 Type II adhesive shall be an elastomeric material with a chlorofluorocarbon propellant and suitable non-flammable solvent as required

3.1.3 Propellant

3.1.3.1 Type I propellant shall be hydrocarbon or blend of hydrocarbons, including halogenated hydrocarbons, as required, to assure that the adhesive conforms to the requirements specified herein.

3 1 3.2 Type II propellant shall be a chlorofluorocarbon or blend of chlorofluoro carbons, as required, to assure that the adhesive conforms to the requirements specified herein.

3.2 Dispenser

3.2.1 The dispenser shall be a commercial type metal pressure container generally known as an aerosol dispenser of such construction as to assure acceptance of the finished packaged by common carriers operating in interstate commerce The capacity of the container shall be sufficient to contain the specified net contents of the product with outage in accordance with generally recognized safe commercial practice Each container shall be labeled as specified in 5.3.

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3.2.2 Dispensing valve. The valve shall have a spray head which can be removed without releasing pressure from the aerosol. The spray head shall have a metering orifice accessible to cleaning with a common pin after removal from the container. The removable spray head shall contain orifices of such dimensions as to produce spraying properties as required in 3.4.2.4. The lower end of the dip tube shall not elongate and touch the bottom of the can and shall have a clearance of one-quarter inch to three-quarters inch from the bottom rim of the can after storage for one week in the product (which allows for swelling of the tube).

3.2.3 Cover can The valve shall be protected from accidental functioning and damage by a press fitting plastic cover cap which shall be removable and replaceable.

3.2.4 Agitation The adhesive shall be so formulated and the dispenser so constructed that mild agitation of the dispenser by hand will ensure that the adhesive and dispenser conform to all related physical requirements specified herein (see 3.3.1.2).

3.3 Qualitative requirements.

3.3.1 Adhesive

3.3.1.1 Color. A sprayed dry film of the adhesive shall be translucent and shall be not darker than a light cream in color.

3.3.1.2 Condition in container. The adhesive shall be easily dispersed when the container is shaken normally by hand (See 3.2.4).

3.3.1.3 Storage stability. The adhesive, with its propellant, shall conform to paragraphs 3.3.1.1, 3.3.1.2, 3.3.2.1, 3.4.2.3, and 3.4.2.4 of this specification when stored and ready for use in the dispenser specified herein for a period of not less than six months at 73.4° ± 2°F. (23° ± 1.1°C). The supplier shall certify that the products furnished meet the specified storage stability requirements.

3.3.2 Dispenser package

3.3.2.1 Valve operation. The valve shall operate freely without excessive finger pressure, shall close immediately upon release of finger pressure, shall operate in such manner that spray is continuous without sputtering or interruption, and shall not clog nor collect heavy deposits of adhesive coating around the orifice

3.4 Quantitative requirements

3.4.1 Adhesive.

3.4.1.1 Total solids. When tested as specified in 4.4.1, the total solids shall be not less than 20 percent for type I and not less than 8 percent for type II of the total net contents of the adhesive dispensers by weight

3.4.1.2 Adhesion. The adhesive shall have a peel adhesion strength of not less than 5 pounds per inch width when tested as specified in 4.4.2.

3.4.2 Dispenser package.

3.4.2.1 Net contents. Type I adhesive shall be furnished in dispensers of two sizes having nominal net contents of 24, 20, and 16 fluid ounces, and type II adhesive shall be furnished in dispensers having nominal net contents of 16 fluid ounces When tested (for weight only) as specified in 4.4.3, the actual net contents shall be as follows:

	Type I 24-ounce can	Type I 20-ounce can	Type I 16-ounce can	Type II 16-ounce can
By weight (ounces)	17.0 to 17.5	14.0 to 14	11.2 to 11.7	20.2 to 20.9
By weight (grams)	483 to 498	397 to 411	312 to 326	547 to 594
By volume(fluid ounces)	23.0 to 23.7	17 to 17.5	15.0 to 15.5	15.1 to 15.7
By volume (cubic centimeters)	680 to 701	503 to 520	444 to 458	447 to 464

3.4.2.2 Internal pressure. When tested as specified 4.4.4, the internal pressure, in pounds per square inch, gage, (p.s.i.g.) shall be as follows:

Type I - 25 to 46 at 73.4°F. (23°C), and not more than 90 at 130°F. (54.4°C)
Type II- 30 to 55 at 73.4°F (23°C), and not more than 110 at 130°F. (54.4°C).

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3.4 2.3 Delivery rate When tested as specified in 4 4 5, the dispenser shall spray the adhesive as follows:

Type I - not less than 20 and not more than [illegible] grams per minute (g. per min.)
II - not less than 55 and not more than [illegible] grams per minute (g. per min.).

3.4.2.4 Spraying properties When tested as specified in 4 4.6, the dispenser shall deliver a spray pattern approximately one inch wide for type I and three inches wide for type II when held at a distance of five to ten inches from a vertical surface and spraying is done at 73.4°F. ± 2.0°F. (23° ± 1.1°C.). The spray pattern so deposited shall be uniform on the smooth vertical surface specified. The adhesive must retain the spray pattern requirements through 90 percent exhaustion of the container.

3.4.2.5 Non-flammability (type II only) There shall be no flame projection when the contents of the pressurized container are tested as specified in 4 4 7.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. Sampling for visual examination and testing shall be in accordance with MIL-STD-105.

4.2.1 Visual examination. At inspection level S-4, with an acceptable quality level (AQL) of 2.5 defects per one hundred units, check for the following: Leaks, rust, damages (dents, bruises, scratches), film defects, improper marking, ease of operation (shake and spray), and other deficiencies.

4.2.2 Tests Conduct tests at inspection level S-3 with an QPL of 2.5 defects per one hundred units for each test performed on end items (filled and pressurized dispensers). Failure to pass a test shall constitute a defect

4.3 Inspection. Inspection shall be performed in accordance with methods 1011 and 1031 of Fed. Test Method Std No 141

4.3.1 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5 Defects shall be scored in accordance with table I. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an QPL of 4.0 defects per hundred units.

TABLE I. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect, illegible; improper size, location, sequence, or method of application
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.
Contents (exterior and interior container)	Number per container is more or less than required. Net weight exceeds requirements

4.4 Test procedures.

4.4.1 Total solids. Test for total solids by the CSMA December 4, 1956, Vacuum Distillation Method for the Determination of Volatile - Nonvolatile Ratios of Aerosol Formulations

4.4.2 Adhesion

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4.4.2.1 Equipment

(a) Galvanized steel panels, 4 inches by 3 inches, 0 inch nominal thickness, conforming to type I, class C or E, of QQ-S-775
(b) Number 10 cotton strips, 1-1/2 inches by 12 inches, conforming to ASTM D 230-44.
(c) Scott tensile tester, or equivalent, with a jaw separation rate of 2 inches per minute.
(d) A steel roller 2 inches wide and weighing approximately 650 grams, capable of applying pressure by hand to the test assembly.
(e) Timing device such as a stopwatch.

4.4.2.2 Procedure Have the galvanized panels thoroughly degreased, clean, and dry Attach two of these panels to a vertical or almost vertical surface in a suitable manner so that adhesive may be applied from the pressurized dispenser evenly and conveniently. Attach four of the cotton duck strips with the weave (warp) in the 12-inch direction in a similar manner to the same surface. Apply three even coats of adhesive (enough to fill in the weave) to the canvas strips and two even coats to the galvanized panels (at the time the second and third coats are applied to the cloth). Allow a drying time of 15 minutes after application of each coat of adhesive. Five minutes after applying the last coats, bond two of the cotton strips to each of the galvanized panels by placing the strips, with adhesive coat facing down, to the panels with the adhesive-coated side turned upward. Bond the strips to the panels securely and smoothly by rolling the bonds vigorously with the steel roller. Dry the bonds for 24 hours at 73.4° ± 2°F. (23° ± 1.1°C), and then immediately for 2 days at 120° ± 5°F. (49° ± 3°C) With no further drying time, cut strips one inch wide from the center of the 1-1/2 inch bonded pieces Pull each of the four bonds thus prepared in 180° peel on the tensile testing machine at a jaw separation rate of 2 inches per minute Record the peel adhesion value for each of the four specimens and report the average of the four values in pounds per inch width for determining conformance with 3.4.1.2

4.4.3 Net contents. Weigh the full, pressurized, ready-to-use dispenser of adhesive accurately to the nearest gram. Then, with the dispenser well agitated and in an upright position and using a needle point, pierce the top with a very fine hole which will allow the propellant to escape. (Conditioning the dispenser for one hour at a relatively low temperature will facilitate this.) After a few minutes the pierced hole can be enlarged and, as soon as there is no further evidence of gas escaping, the top of the dispenser should be cut open. Thoroughly clean all interior surfaces of the dispenser and top by successive washing with hexane or toluene. Dry the clean, empty dispenser and top, and weigh accurately to the nearest gram Determine the net contents by calculating the difference between the full, ready to use dispenser and the clean, empty dispenser. All tests for net contents shall be made on the same dispenser, and not less than two dispensers of any one or both sizes from the same lot shall be tested for conformance with 3.4.2.1 Report the average of the results obtained on the two or more dispensers tested.

4.4.4 Internal pressure

4.4.4.1 Equipment. U.S. Gauge Company gage number 10897, or equivalent (calibrated to read 0 to 100 p.s.i.g), with valve fitting for insertion into pressurized aerosol dispensers with a Sprayon Danvern valve, or equal

4.4.4.2 Procedure Condition the pressurized dispensers to be tested at the temperature specified in 3.4.2.2. Turn the dispenser upside down with the spray tip seated correctly in the valve. Clean the valve of adhesive by depressing the spray tip until only gas is emitted from the tip. Keeping the dispenser in the inverted position, push the gage into the valve and immediately read the gage. Pull the gage from the dispenser before turning the dispenser to the upright position. Not less than six dispensers shall be tested at each specified temperature from an inspection lot. Report the maximum, minimum, and average pressures for each specified temperature for conformance with 3.4.2.2

4.4.5 Delivery rate

4.4.5.1 Equipment

(a) Balance or scale accurate to ± 1 g.
(b) Timing device such as a stopwatch.
(c) Spray booth or area in which adhesive can be sprayed safely for one minute continuously

1/ A roller similar to the one specified may be purchased from the Huggson and Pettis Manufacturing Co., Orange, Connecticut

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4.4.5.2 Procedure. Condition the pressurized dispenser at 73.4° ± 2°F (23° ± 1.1°C). Weigh the conditioned pressurized dispenser. Then, holding the dispenser in a vertical position, depress the valve activator all the way down, and spray the adhesive at full valve opening for 60 seconds. Reweigh the dispenser and calculate the delivery rate from the loss in weight from the original. Report the average delivery rate of six dispensers from each inspection lot for conformance with 3.4.2.5.

4.4.6 Spraying properties. Condition the adhesive dispenser for one hour at 73.4° ± 2°F (23° ± 1.1°C). Use No. 1 kraft, wrapping paper, untreated, 90 pounds basis weight (24 inches by 36 inches, 500 sheets). Attach a sheet of this paper, approximately one yard square, in a vertical position to a wall or other stationary surface. Holding the pressurized dispenser about five to ten inches from the surface, spray the adhesive on the kraft paper for approximately ten seconds without moving the dispenser. Observe the adhesive pattern thus sprayed on the paper surface. If for any reason the spraying is not done in a manner so that an even, continuous flow of adhesive is delivered, the dispenser should be agitated and the test repeated. Not less than six dispensers of adhesive from each inspection lot shall be tested for conformance with 3.4.2.4.

4.4.7 Non-flammability (type II only). The contents of the pressurized container shall be projected through a flame as described in CSMA Aerosol Guide, September 12, 1963, Revised Flammability Test Methods for Aerosol Products.

5. PREPARATION FOR DELIVERY

5.1 **Packaging.** Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified (see 6.2), 12 filled cans of adhesive of like description, shall be packaged in a close-fitting box conforming to PPP-B-636, class weather-resistant. Fiberboard separators shall be provided with the fiberboard meeting the requirements of PPP-F-320, type CF, class domestic, variety SW. The box shall be closed and sealed in accordance with the appendix of the box specification.

5.1.2 Level B. Level B packaging shall conform with the requirements of 5.1.1, except the box shall conform with PPP-B-636, class domestic.

5.1.3 Level C. The filled cans of adhesive shall be packaged to afford adequate protection against corrosion and damage during shipment from the supply source to the first receiving activity. The suppliers commercial practice will be acceptable when it fulfills the above requirements.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Unit packages of adhesive, of like description, shall be packed in a close-fitting box conforming to PPP-B-585, class 3, PPP-B-591, class II; PPP-B-601, overseas type; PPP-B-621, class 2; PPP-B-636, class weather-resistant, or PPP-B-640, class 2. The boxes shall be closed and strapped in accordance with the appendix of the applicable box specification. The gross weight of the triple-wall fiberboard box and the wood boxes shall not exceed 200 pounds. The gross weight of the PPP-B-636 box shall not exceed the weight limitations of the box specification.

5.2.2 Level B. Filled cans packaged as specified in 5.1.2 shall not require overpacking.

5.2.3 Level C. The filled cans of adhesive shall be packed to insure carrier acceptance and safe delivery at the lowest rates, in containers complying with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.3 Special markings. In addition to other markings specified (see 5.3.1 and 5.3.2), the following markings shall be applied.

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5.3.3.1 Unit containers.

(a) Net weight in ounces.
(b) Name of material
(c) Manufacturer's name and address.
(d) Precautions for use and storage
(e) Directions for use including surface preparation
(f) Date of manufacture by month and year
(g) Date of first re-inspection (12 months after date of manufacture)
(h) Store below 80°F.
(i) Name of the propellant or propellants.

5.3.3.2 Intermediate and shipping containers.

(a) Net weight in ounces.
(b) Name of material.
(c) Manufacturer's name and address
(d) Precuations for use and storage.
(e) Directions for use including surface preparation.
(f) Date of manufacture by month and year.
(g) Date of first re-inspection (12 months after date of manufacture)
(h) Store below 80°F.
(i) Name of the propellant or propellants

6. NOTES

6.1 Intended use. This adhesive in pressurized dispensers is intended to provide a quick, convenient method for bonding a wide variety of adherends. It will bond such materials as paper, glass, cardboard, cloth, wood, plastic films, metal, and foam rubber to themselves, to one another or to other materials. It is not intended for structural use or for bonding certain equipment components where such critical properties as high pressure or dielectric strength may be required.

6.1.1 Type I adhesive has the advantage of higher solids and lower operating cost than type II, but type II adhesive should be used when safety precautions are necessary

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification
(b) Type of adhesive.
(c) Nominal size dispenser selected (see 3.4.2.1)
(d) Selection of applicable levels of packaging and packing required (see 5.1 and 5.2).
(e) Quantity required
(f) Special marking required (see 5.3.3).

MILITARY INTERESTS:

Custodian:

Navy - AS

Coordinating Activity

Navy - AS

User Activities

Army - MR
Navy - SH

Preparing Activity:

GSA - FSS

CIVIL AGENCY COORDINATING ACTIVITY:

COM - NBS

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